

Victor Chua

Singapore | +65 9190 5651 | chua1102@e.ntu.edu.sg | github.com/zeppyclown
Singaporean Citizen | [linkedin.com/in/vicchua123](https://www.linkedin.com/in/vicchua123) | Internship: January – August 2027 | Open to remote
Applied AI Engineering | Deterministic Scoring + LLM Systems | Evaluation & Guardrails

Education

Nanyang Technological University (NTU) Aug 2024 – Dec 2028
Bachelor of Computing (Computer Science) with Turing AI Scholars Programme (TAISP)

Work Experience

SUSS Digital Solutions & AI Singapore
AI & Software Engineer Intern May 2026 – Aug 2026
Lecture-Audit Pipelines - Speech Measurement & Evaluation Layers on a Team Codebase

- Replaced LLM-graded lecture scoring with a **deterministic 7-factor formula** over locally measured acoustics (**faster-whisper** + Praat); benchmarked GPT-4o on that formula at 7.06 MAE and kept it out of the scoring path
- Stopped ASR word-confidence driving an articulation score once it penalised accented English; it now ships labelled **"ASR intelligibility, not speech quality"**, with real intelligibility a human-entered field
- Built an offline **eval harness** over stored audit reports (**1,529 assertions** in about a second, no API spend), replaying scoring guardrails as output invariants, with 22 fault-injection tests proving each check fails
- Reported LLM-judge agreement on **held-out** modules separately once grounding the judge with four worked examples had contaminated four of eleven, so the headline figure was not a blind test and was labelled as such

Yap Motor Singapore
Machine Learning Intern Dec 2025 – Jan 2026
First Production AI System: Retrieval-Augmented Generation

- Built Yap Motor's first AI system, a production **RAG** pipeline (**Google Gemini** + **ChromaDB**) letting 24 mechanics query 500+ pages of multilingual repair manuals; still in use 6 months after the internship
- Reached **90% retrieval precision** across 8 diagnostic categories by iterating on chunk size, overlap, and embedding strategy, validated against a month of structured mechanic feedback on live repair queries
- Engineered a multi-stage retrieval pipeline: folder-based priority scoring, **regex NLP** pulling vehicle make/model/year from natural language, dynamic ChromaDB filters, and **re-ranking** before LLM reasoning
- Built a multilingual (EN/CN/JP) ingestion pipeline for PDFs and Word docs of variable structure; debugged two production blockers: ChromaDB schema validation failures and case-sensitive metadata filters

DBS Singapore
Data Engineer Intern May 2023 – May 2024
Enterprise Pipeline Monitoring & ML Feature Retrieval

- Built a **Grafana/SQL** dashboard tracking RAM and execution metrics across every version of a 3,000+ job/day pipeline; surfaced version-specific bottlenecks that teams then fixed for 20% RAM and 15% speed gains
- Developed a **REST client** integrating **Elasticsearch** into the team's internal Python library, giving production ML models real-time feature retrieval through shared infrastructure for the first time

Personal Projects

Drum Hub – AI drum-score reader | hf.co/Icyautumn/drum-omr-mobilenetv3 Jun 2026 – Present

- Trained a dual-head **MobileNetV3** optical music recognition model (2.47M params) by two-phase transfer learning: 90.9% duration accuracy, 0.93/0.89 kick/snare F1 on 1,924 held-out bars from 26 unseen songs
- Exported the trained **PyTorch** model to **ONNX** and published the weights and ONNX artifact under MIT, so the score reader runs offline on a student's machine with no network call or per-use API cost
- Built the score editor in **TypeScript** with **VexFlow 5** inside an **Electron** desktop app: correct voice-split stem direction logic and note-level editing from rest through 32nd notes with beam linkages

Leadership & Activities

TAISP x MLDA Workshop Lead, NTU School of Computing 2025
Designed and delivered a hands-on PyTorch workshop curriculum with teaching notebooks for NTU undergraduates, taking students from tensor basics to a trained image classifier in a single session.

Technical Skills

Languages: Python, TypeScript, SQL, JavaScript

AI & LLM Systems: RAG, ChromaDB, Vector Search, Re-ranking, On-Premises Inference (local 7B, Ollama), LLM APIs (Gemini, GPT-4o, Claude), Prompt Design, Structured Output

Evaluation & Reliability: Eval Harness Design, Fault-Injection Testing, Output Invariants, Held-Out Agreement, Scoring Guardrails, Human-in-the-Loop Gates

Speech & ML: faster-whisper, Praat / parselmouth, PyTorch, Transfer Learning, MobileNetV3, ONNX Runtime, HuggingFace

Engineering & Data: Electron, VexFlow 5, REST API Design, Elasticsearch, Grafana, Git (version control), Vector & Relational Databases